

Question 01

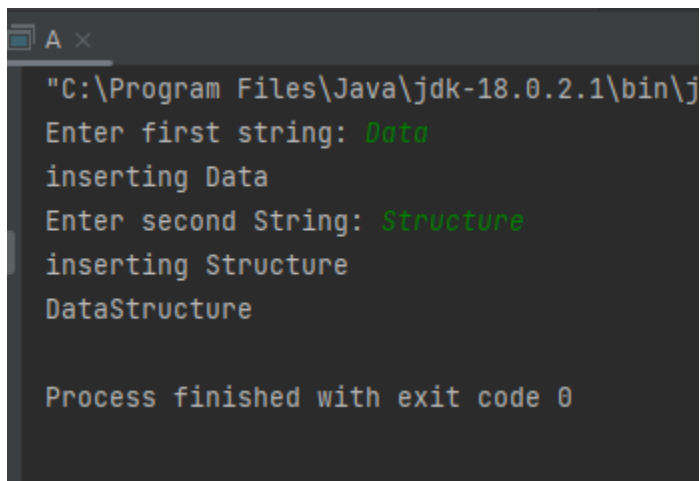
```
public class Queue_String {
    private int front;
    private int rear;
    private int maxSize;
    private int count;
    String[] arr;
    Queue_String(int max){
        maxSize=max;
        arr=new String[maxSize];
        front=0;
        rear=-1;
        count=0;
    }
    public boolean isEmpty() {
        if(rear<front){
            return true;
        }else{
            return false;
        }
    }
    public boolean isQueueFull() {
        if(rear==maxSize-1){
            return true;
        }else{
            return false;
        }
    }
    public void append(String item){
        if(isQueueFull()){
            System.out.println("\nQueue is full\n");
        }else{
            System.out.println("inserting "+item);
            arr[++rear]=item;
            count++;
        }
    }
    public String serve(){
        if(isQueueEmpty()){
            System.out.println("\nQueue is empty.\n");
            return "";
        }else{
            String item=arr[front++];
            count--;
            return item;
        }
    }
}
```

```
import java.util.Scanner;

public class A {
    public static void main(String[] args) {
        Queue_String queue=new Queue_String(2);
        Scanner scn=new Scanner(System.in);
        System.out.print("Enter first string: ");
        queue.appended(scn.next());

        System.out.print("Enter second String: ");
        queue.appended(scn.next());

        StringBuilder newText= new StringBuilder(queue.serve());
        newText.append(queue.serve());
        System.out.println(newText);
    }
}
```



```
"C:\Program Files\Java\jdk-18.0.2.1\bin\j
Enter first string: Data
inserting Data
Enter second String: Structure
inserting Structure
DataStructure

Process finished with exit code 0
```

Question 02

```
public class Queue_Char {
    private int front;
    private int rear;
    private int maxSize;
    private int count;
    char[] arr;
    Queue_Char(int max){
        maxSize=max;
        arr=new char[maxSize];
        front=0;
        rear=-1;
        count=0;
    }
    public boolean isEmpty(){
        if(rear<front){
            return true;
        }else{
            return false;
        }
    }
}
```

```

    }
}
public boolean isQueueFull(){
    if(rear==maxSize-1){
        return true;
    }else{
        return false;
    }
}
public void append(char item){
    if(isQueueFull()){
        System.out.println("\nQueue is full\n");
    }else{
        System.out.println("inserting "+item);
        arr[++rear]=item;
        count++;
    }
}
public char serve(){
    if(isQueueEmpty()){
        System.out.println("\nQueue is empty.\n");
        return '0';
    }else{
        char item=arr[front++];
        count--;
        return item;
    }
}
}

```

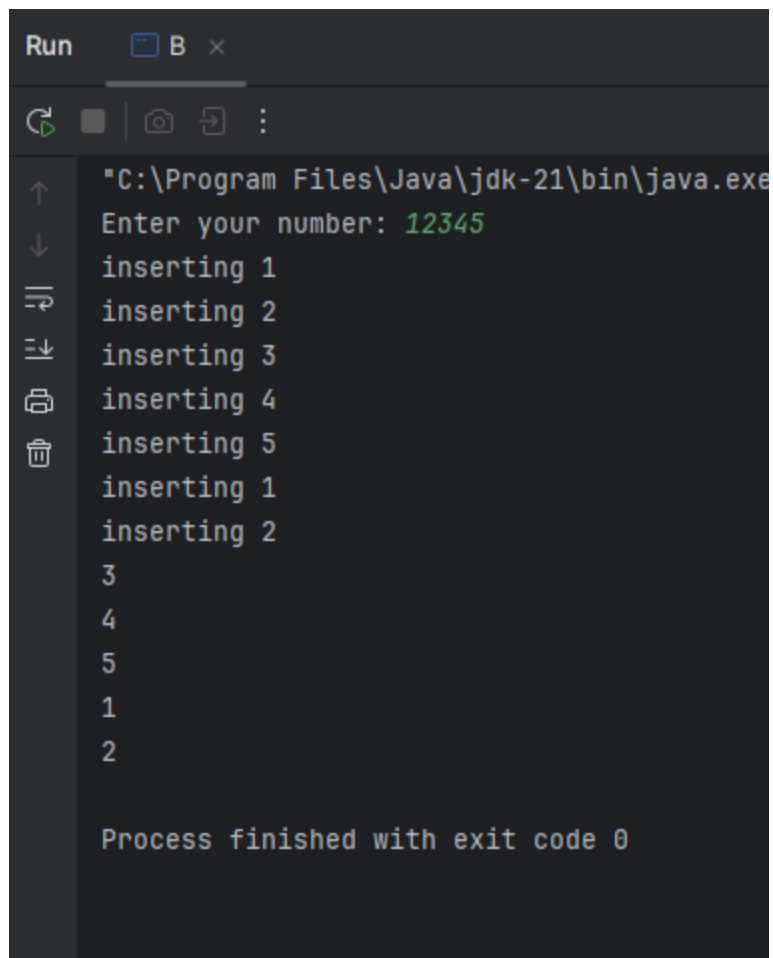
```

import java.util.Scanner;

public class B {
    public static void main(String[] args) {
        Queue_Char queue=new Queue_Char(20);
        Scanner scn =new Scanner(System.in);
        System.out.print("Enter your number: ");
        String number=scn.next();
        int d=number.length()/2;
        char[] textArr=number.toCharArray();
        for(char x:textArr){
            queue.append(x);
        }
        for(int i=0;i<d;i++){
            char c=queue.serve();
            queue.append(c);
        }
        while (!(queue.isQueueEmpty())){
            System.out.println(queue.serve());
        }
    }
}

```

```
}
```



```
Run B x
"C:\Program Files\Java\jdk-21\bin\java.exe"
Enter your number: 12345
inserting 1
inserting 2
inserting 3
inserting 4
inserting 5
inserting 1
inserting 2
3
4
5
1
2
Process finished with exit code 0
```

Question 03

same queue class that we used in second question can be used for this question.

```
import java.util.Locale;
import java.util.Scanner;

public class C {
    public static void main(String[] args) {
        Queue_Char queue=new Queue_Char(5);
        Scanner scn=new Scanner(System.in);
        queue.append('A');
        queue.append('E');
        queue.append('I');
        queue.append('O');
        queue.append('U');

        System.out.print("Enter your word: ");
```

```
char[] text=scn.next().toUpperCase(Locale.ROOT).toCharArray();
char vowel;
while (!queue.isEmpty()){
    vowel= queue.serve();
    for (char x:text){
        if(x == vowel){
            System.out.println(x);
        }
    }
}

}
```

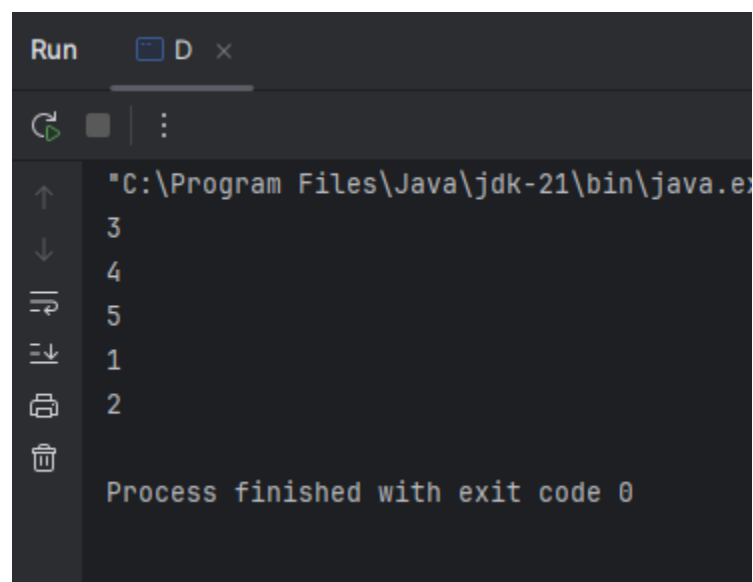
```
↑ "C:\Program Files\Java\jdk-18.0.2.1\bin\java.exe
↓ inserting A
↻ inserting E
⏮ inserting I
⏪ inserting O
⏩ inserting U
🗑 Enter your word: DIALOGUE
A
E
I
O
U

Process finished with exit code 0
```

Question 04

```
public class Node_int {  
    int data;  
    Node_int next;  
}
```

```
public class D {  
    public static void main(String[] args) {  
        Queue_int queue=new Queue_int();  
        for(int i=1;i<=5;i++){  
            insertElement(i,queue);  
        }  
        rotate(2,queue);  
  
        while (!(queue.isQueueEmpty())){  
            System.out.println(queue.serve());  
        }  
    }  
    public static void insertElement(int x,Queue_int q){  
        q.append(x);  
    }  
    public static void rotate(int r,Queue_int q){  
        for (int j=0;j<r;j++){  
            int n=q.serve();  
            q.append(n);  
        }  
    }  
}
```



```
Run D x  
C:\Program Files\Java\jdk-21\bin\java.exe  
3  
4  
5  
1  
2  
Process finished with exit code 0
```

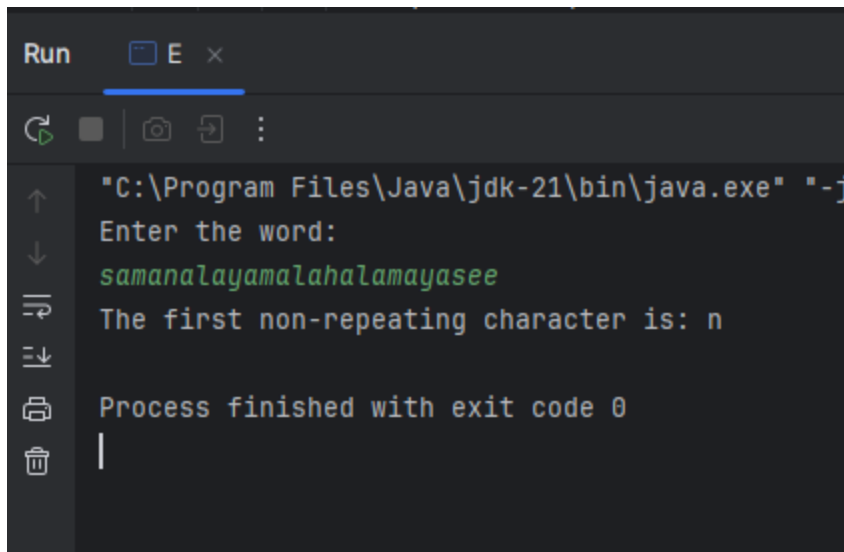
Question 05

```
public class Node_char {  
    char data;  
    Node_char next;  
}
```

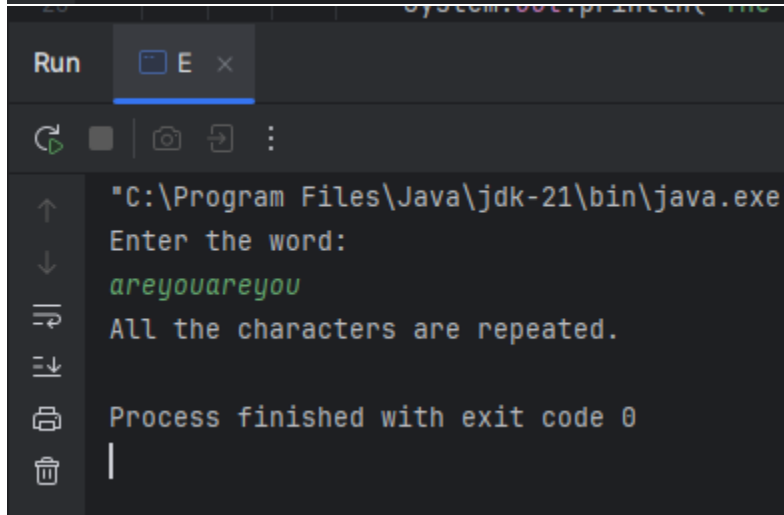
```
public class Queue_char_linked {  
    Node_char front;  
    Node_char rear;  
    int queueSize;  
    Queue_char_linked(){  
        front=null;  
        rear =null;  
        queueSize=0;  
    }  
    public boolean isEmpty(){  
        if(queueSize==0){  
            return true;  
        }else{  
            return false;  
        }  
    }  
    public char serve(){  
        if (isEmpty()){  
            System.out.println("Queue is empty.");  
            return 0;  
        }else{  
            char n=this.front.data;  
            this.front=front.next;  
            queueSize--;  
            return n;  
        }  
    }  
    public void append(char data){  
        Node_char oldRear=rear;  
        rear=new Node_char();  
        rear.data=data;  
        rear.next=null;  
        if(isEmpty()){  
            front=rear;  
        }else{  
            oldRear.next=rear;  
        }  
        queueSize++;  
    }  
}
```

```
import java.util.Scanner;

public class E {
    public static void main(String[] args) {
        Scanner scn=new Scanner(System.in);
        Queue_char_linked queue=new Queue_char_linked();
        System.out.println("Enter the word:");
        String text=scn.nextLine();
        char[] textArr=text.toCharArray();
        for(char x:textArr){
            queue.append(x);
        }
        char checkLetter;
        boolean allRepeat=true;
        outer:
        while(!(queue.isEmpty())){
            checkLetter= queue.serve();
            int c=0;
            for(char x:textArr){
                if(checkLetter==x){
                    c++;
                }
            }
            if(c<2){
                System.out.println("The first non-repeating character is: "+checkLetter);
                allRepeat=false;
                break outer;
            }
        }
        if(allRepeat){
            System.out.println("All the characters are repeated.");
        }
    }
}
```

```
Run E x
Enter the word:
samanalayamalahalamayasee
The first non-repeating character is: n
Process finished with exit code 0
```



```
Run E x
Enter the word:
areyouareyou
All the characters are repeated.
Process finished with exit code 0
```